Portaldot Hacker House 2026 · Multi-Chain & Cross-Chain Infra

Kurs

Price feeds for Portaldot V3, signed by independent reporters, readable from Solidity and ink!, portable to any EVM chain.

Kurs: Indonesian for exchange rate, the number on every money-changer board.

Portaldot already has money apps. None of them can read a price.

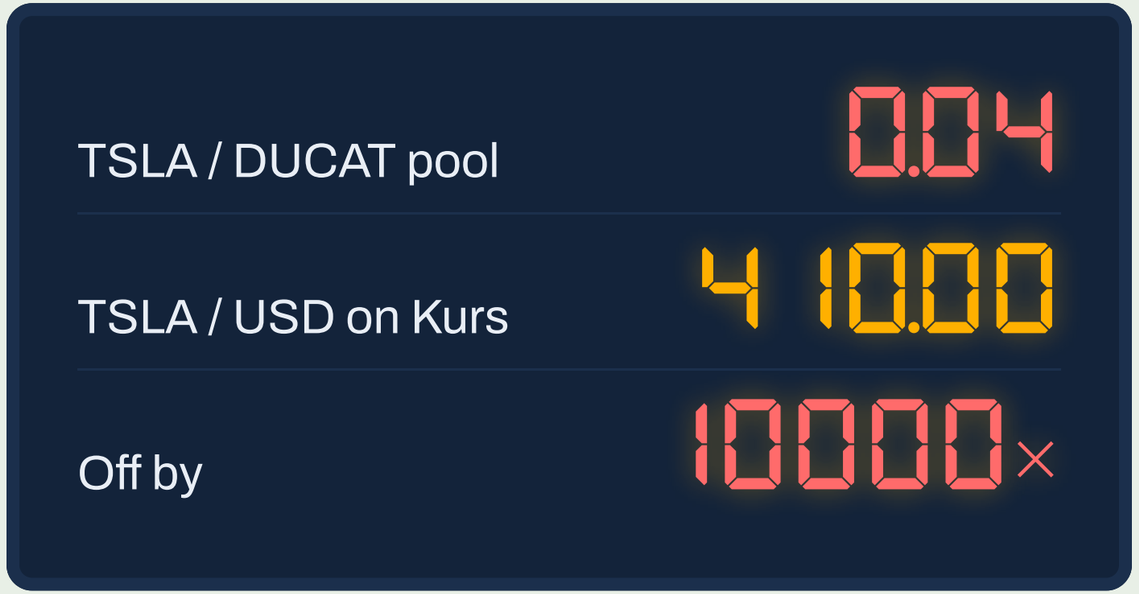
Checked on V3 testnet, spec 3002	Found
Oracle pallet (index 85)	**0 values**
Oracle operator set	**empty**
Oracle precompile for EVM	**none**
Chainlink, Pyth, RedStone	**not deployed**
Tokenised stocks on the native DEX	TSLA, NVDA, AAPL, AMZN, META
Hacker-house lending, stablecoin, escrow apps	a dozen and counting

Ask the pools what TSLA is worth

Route one TSLA token through Portaldot's own pools (TSLA → DUCAT → tPOT → pUSD) and you get **about $0.04**.

Kurs says **about $410**.

A lending market that prices collateral from these pools can be drained by anyone with a few swaps. That is how most DeFi exploits start.



How Kurs works

1

Read

Reporters read exchanges, aggregators, Chainlink on Ethereum, FX and equity quotes, and Portaldot's own pools.

2

Sign

Each reporter signs its own number with EIP-712. No chain id in the domain.

3

Relay

Anyone submits the bundle. The sender is not trusted; the signatures are.

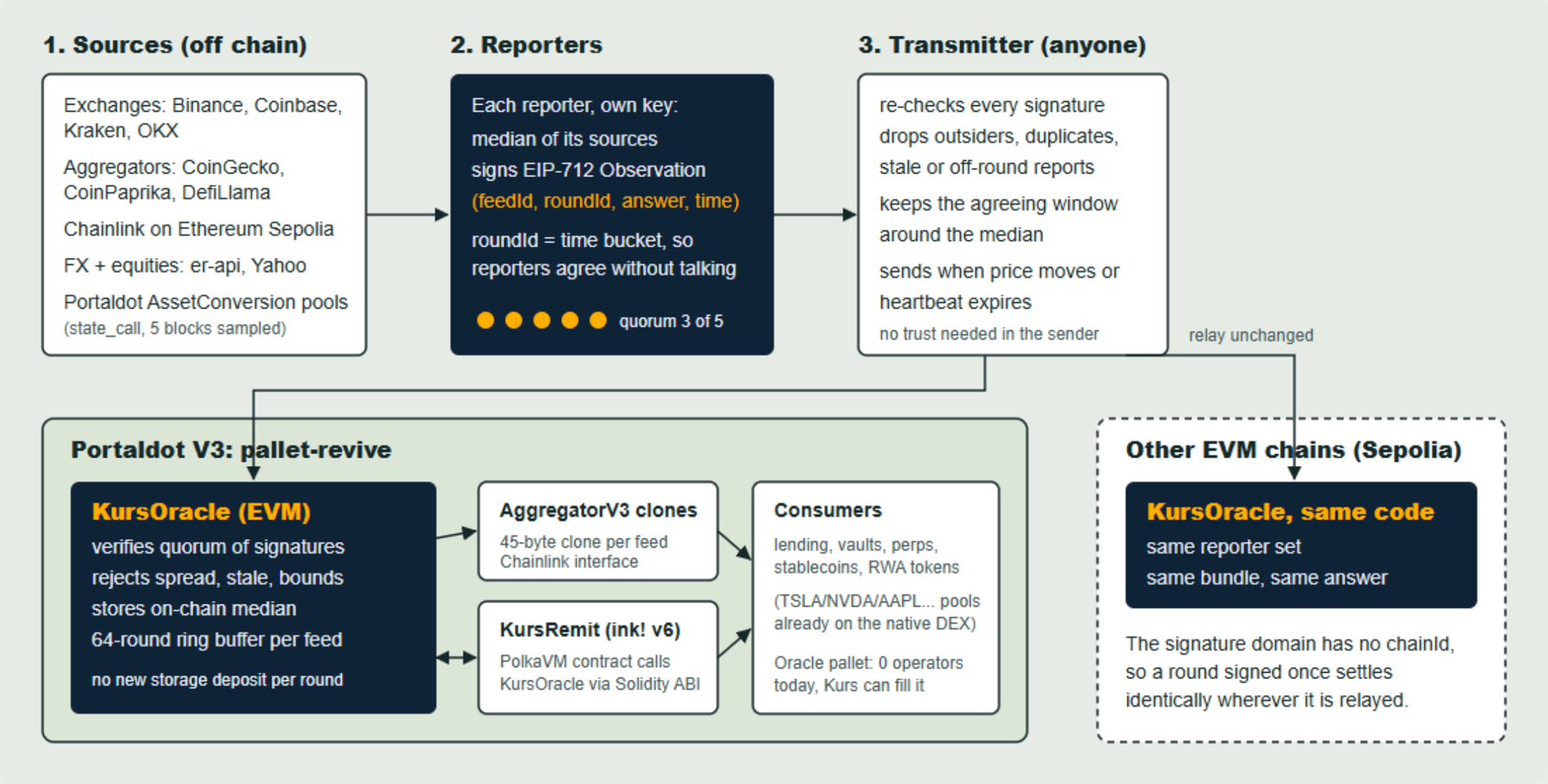
4

Settle

KursOracle checks a majority quorum, freshness, spread and bounds, and stores the median.

A minority of dishonest reporters cannot move the median, and cannot forge a round.

One oracle, both VMs, every chain



Why this has to be Portaldot

Dual VM in one pallet. Solidity reads Kurs through the Chainlink interface. ink! v6 reads the same contract through a Solidity-ABI call. One oracle, not two.

A native price to export. tPOT/pUSD comes from `pallet-asset-conversion` through a runtime API, sampled over 20 minutes of blocks, then relayed to Ethereum with the same signatures.

A slot already waiting. The runtime ships an Oracle pallet with no operators. Kurs reporters are ready to fill it, so native pallets and contracts read the same prices.

Built for Revive's costs. A 64-round ring buffer and 45-byte clones mean a new round adds no storage deposit.

Built during the hacker house

11

feeds: BTC, ETH, DOT, USDT, USD/IDR, five equities, tPOT

87

tests: 42 Solidity incl. fuzz and invariants, 17 TypeScript incl. anvil e2e, 28 Playwright incl. accessibility and layout shift; Slither in CI

2

VMs reading one oracle: EVM and PolkaVM (ink! v6)

5

reporters, quorum 3, median on chain

0

servers: the dashboard reads public RPCs in the browser

2

ways in: pushed by the relay, or pulled into your own transaction with ERC-7412

Proof on chain

What	Where
KursOracle v2, factory, 11 feed clones, KursConverter	Ethereum Sepolia, oracle `0xd3f6…cbc3`, all verified on Sourcify
Rounds pushed by 5 reporters, every signer on chain	RoundUpdated events, re-verified in the browser by the dashboard
ERC-7412 pull in one transaction	Multicall3: fulfillOracleQuery + convert, tx `0x9e2a…405f`
ink! v6 reads the Solidity oracle	pallet-revive node: KursRemit.latest() equals the oracle round
Portaldot-native price on Ethereum	tPOT/pUSD from AssetConversion, live on Sepolia
Live on Portaldot V3	KursOracle `0x5bdd…56f4`, 11 feeds, rounds every ~64 min at ~0.1 tPOT, mirrored to Sepolia
Owned by a native Portaldot multisig	2-of-3 pallet-multisig; acceptOwnership executed via asMulti + revive.call
ERC-7412 pull on Portaldot	fulfillOracleQuery then convert, $100 to Rp1,814,489.94

Every address and transaction hash:

`github.com/PugarHuda/kurs/blob/main/docs/evidence.md` ·

live: `kurs-ten-zeta.vercel.app`

Who pays, and why

Users: every Portaldot app that moves value against a price. Lending (DotProof), stablecoins (Ballast, StableTap, pUSD), tokenised stocks (PASTX pools), escrow and payroll in IDR.

Feed sponsorship: an app pays a monthly POT fee per feed it needs at a given heartbeat. Sponsors fund reporter rewards and transmit gas.

Export fees: Portaldot-native prices (tPOT, ecosystem tokens) relayed to Ethereum, BNB and Base for bridges and CEX listings.

Reporter market: community operators stake POT to join the set and earn from sponsorship. Misbehaviour is visible on chain in every round's signer bitmap.

Next

- Independent reporter operators from the Portaldot community
- Owner key behind Portaldot's native Multisig + Proxy with a delay
- Kurs reporters as operators of the runtime Oracle pallet
- Read precompile for the Oracle pallet
- Relay to Base, BSC, Arbitrum; POT feeds via bridge.portaldot.dev
- Paid equity data before mainnet; external audit

Team

Pugar Huda Mantoro

Built pdk, the Portaldot Dev Kit: 1st place, Portaldot Online Mini Hackathon S1, Builder Tools. Published on PyPI and npm.

Kurs is open source under MIT:
github.com/PugarHuda/kurs

